Code and Branch Coverage Report

Online Exam Portal | Software Engineering Mini-Project, Part 3 | 10 November 2026

Student: Chiranthan V | SRN: PES2UG24CS138 | Section C

Release identifier: sprint2-final-2026-11-10 | Final submission report

Verified application commit: 14074716e3c2de2f59da3077ef265546004e704c

Final main CI revision: 14074716e3c2de2f59da3077ef265546004e704c. Backend, browser and performance runs below executed this merged revision.

The results below refer to these code versions. Earlier operational tests are listed separately with their dates.

1. Measured scope and result

Backend statement coverage: 1,099 / 1,144 = 96.07%. Backend branch coverage: 213 / 252 = 84.52%. The coverage.py combined statement+branch metric is (1,099+213)/(1,144+252) = 93.98%, displayed as 94% by its console report. These metrics must not be interchanged.

Core scope services.py + assessment.py: 65/67 statements = 97.01%; 28/30 branches = 93.33%; combined 95.88%. The configured 70% core combined gate and 85% whole-scope combined gate both passed. No separate 85% branch-only gate is configured; 84.52% branch coverage must not be rounded up to claim that.

2. Instrumentation and denominator

coverage.py 7.17.0, branch=true, executed with 101 PostgreSQL backend tests on the named GitHub merge revision. Configured source is backend/portal. Migrations, tests and management commands are omitted. Django configuration, scripts, frontend JavaScript, third-party dependencies and container files are outside this coverage denominator. Monitor/browser execution results are separate; frontend statement/branch coverage was not collected. The recheck reproduced the original coverage percentages.

A statement means an executable source line counted by coverage.py, not every physical line. Branches are instrumented possible exits/arcs, not a proof of every Boolean combination or MC/DC. Zero-denominator modules are shown N/A in the tables rather than inventing a measured 100% result.

3. Statement coverage by module

Module	Statements	Covered	Missing	Coverage
__init__.py	0	0	0	N/A (empty)
assessment.py	20	19	1	95.00%
assessment_views.py	186	171	15	91.94%
auth_views.py	156	156	0	100.00%
authentication.py	8	8	0	100.00%
models.py	198	198	0	100.00%
notifications.py	11	11	0	100.00%
permissions.py	7	7	0	100.00%
roster.py	81	75	6	92.59%
serializers.py	98	97	1	98.98%
services.py	47	46	1	97.87%
views.py	332	311	21	93.67%

TOTAL: 1,144 statements; 1,099 covered; 45 missing; 96.07% statement coverage. Core engine statement coverage is 65/67 = 97.01%. Coverage is evidence of execution, not completeness of assertions or absence of defects.

Evidence and reproduction

Team-lead acceptance record: https://github.com/chiranthanv/online-exam-portal/pull/61#pullrequestreview-5838992122 . PR #64 approval/review is recorded through chiranthanv; a separate Harkirat/Dwijesh approval is not asserted.

Backend CI: https://github.com/chiranthanv/online-exam-portal/actions/runs/42786113452

Frontend/browser CI: https://github.com/chiranthanv/online-exam-portal/actions/runs/42786113444

Repository evidence: docs/testing/part3/evidence/2026-11-10-release/. The manifest records SHA-256 hashes. Downloaded archives were checked against GitHub's artifact digests before extraction. The selected JSON and text results are preserved without report screenshots. The original 100-test run remains in evidence/2026-11-06/; the earlier 101-test recheck remains in evidence/2026-11-06-recheck/.

Detailed coverage: coverage.json, coverage.xml and backend-coverage-original.zip (includes the interactive htmlcov/index.html report). Backend log lists every test; frontend JSON lists each assertion/scenario and actual status.

Part B - Branch Coverage Report

4. Branch coverage by module

Module	Exits	Covered	Missing	Partial	Coverage
__init__.py	0	0	0	0	N/A (no branches)
assessment.py	10	9	1	1	90.00%
assessment_views.py	64	49	15	13	76.56%
auth_views.py	16	16	0	0	100.00%
authentication.py	2	2	0	0	100.00%
models.py	2	2	0	0	100.00%
notifications.py	4	4	0	0	100.00%
permissions.py	0	0	0	0	N/A (no branches)
roster.py	34	28	6	6	82.35%
serializers.py	26	25	1	1	96.15%
services.py	20	19	1	1	95.00%
views.py	74	59	15	15	79.73%

TOTAL: 252 possible exits, 213 covered, 39 missing. Thirty-seven partially covered branch locations are reported. Partial-branch locations and missing exits are different counts; do not add them together. Core engine branch coverage is 28/30 = 93.33%.

5. Uncovered source and branch exits

Module	Missing executable lines	Missing exits (source -> destination)
assessment.py	18	17->18
assessment_views.py	74, 75, 98, 123, 148, 190, 194, 195, 196, 242, 244, 251, 261, 264, 324	97->98, 122->123, 147->148, 189->190, 193->194, 195->196, 195->197, 210->212, 241->242, 243->244, 246->251, 254->256, 260->261, 263->264, 323->324
roster.py	34, 39, 41, 48, 76, 80	33->34, 38->39, 40->41, 47->48, 75->76, 79->80
serializers.py	155	154->155
services.py	50	49->50
views.py	103, 104, 131, 210, 213, 349, 350, 423, 431, 432, 436, 451, 456, 458, 460, 513, 514, 516, 518, 524, 525	52->57, 128->131, 209->210, 212->213, 303->312, 348->349, 422->423, 435->436, 450->451, 455->456, 457->458, 459->460, 515->516, 517->518, 521->529

The JSON is the authoritative machine-readable list of missing lines and arcs. Negative arc destinations, if present, identify function exits in coverage.py. Missing coverage is not automatically a product defect; it identifies unexercised behavior for focused review.

Examples: assessment.py line 18/exit 17->18 is the explicit persisted-answer selected_option_id=None guard; current unanswered fixtures also exercise the absence-of-answer route. services.py line 50/exit 49->50 is the exam-total overflow rejection. Additional unexercised roster/permission/error paths remain visible above. No synthetic tests were added merely to increase the percentage, and 100% branch coverage is not claimed.

6. Reproduction and audit checks

```
coverage run backend/manage.py test portal.tests --verbosity 2
coverage report
coverage report --include=backend/portal/services.py,backend/portal/assessment.py --fail-under=70
coverage json -o coverage.json
coverage xml
coverage html
```

Run from repository root using requirements-dev.txt, Python3.11, DEBUG=1 and a disposable PostgreSQL16 DATABASE_URL. pyproject.toml supplies branch/source/omit settings. Open htmlcov/index.html from the original backend artifact to inspect highlighted source. Raw CI JSON/XML and the original archive are retained, avoiding reliance on rounded console text.

The earlier local Windows/SQLite check discovered 100 tests: 95 passed and 5 PostgreSQL-specific tests were skipped. That baseline predates the new simultaneous-start test. Its coverage is stored separately and is not included in the fresh 101-test PostgreSQL figures above.